

Homework Classical Physics, Fall 2023

Prof. J. Fritz, Research III Room 56, jfritz@constructor.university.de

Problem Set 01.3 (L01 & L02)

Due on Thursday, 14. May 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - trajectories (4 points)

The position of a particle along a straight line between the times $t = 0.00 \text{ s}$ and $t = 2.00 \text{ s}$ is given by the function $x(t) = a t^3 - b t^2 + c t$ with: $a = 2.50 \text{ m/s}^3$, $b = 10.0 \text{ m/s}^2$, $c = 8.00 \text{ m/s}$.

- a) At what time(s) is the particle (instantaneously) at rest?
- b) At what time(s) is the particle's velocity (instantaneously) not changing?
- c) What is the average velocity of the particle between times $t = 1.20 \text{ s}$ and $t = 1.80 \text{ s}$?